

Health IT Diffusion and Physician Density

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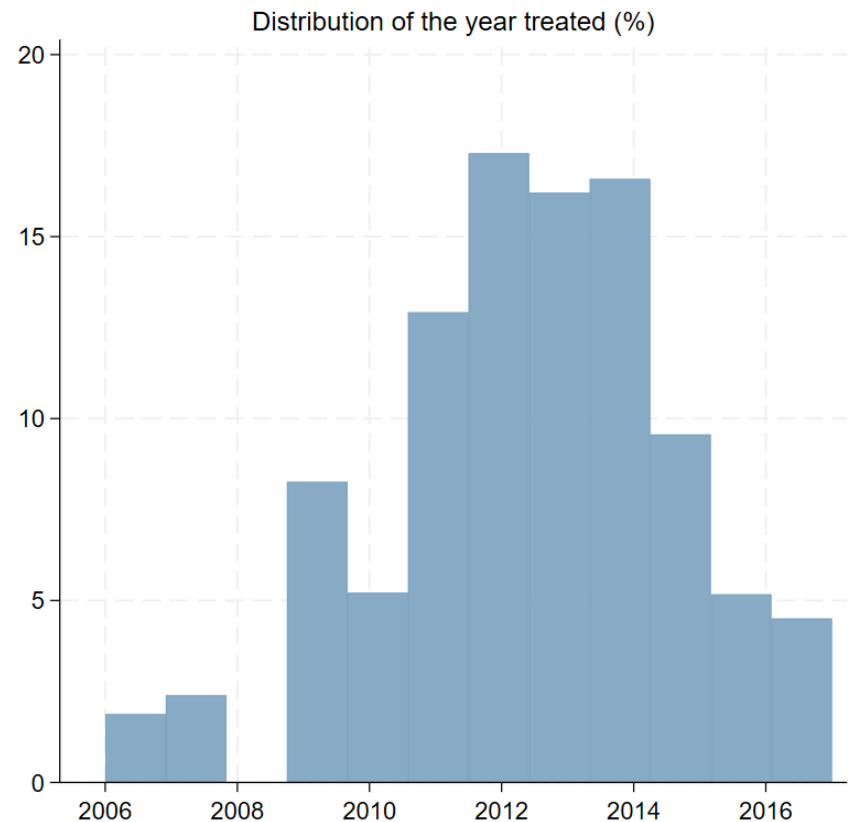
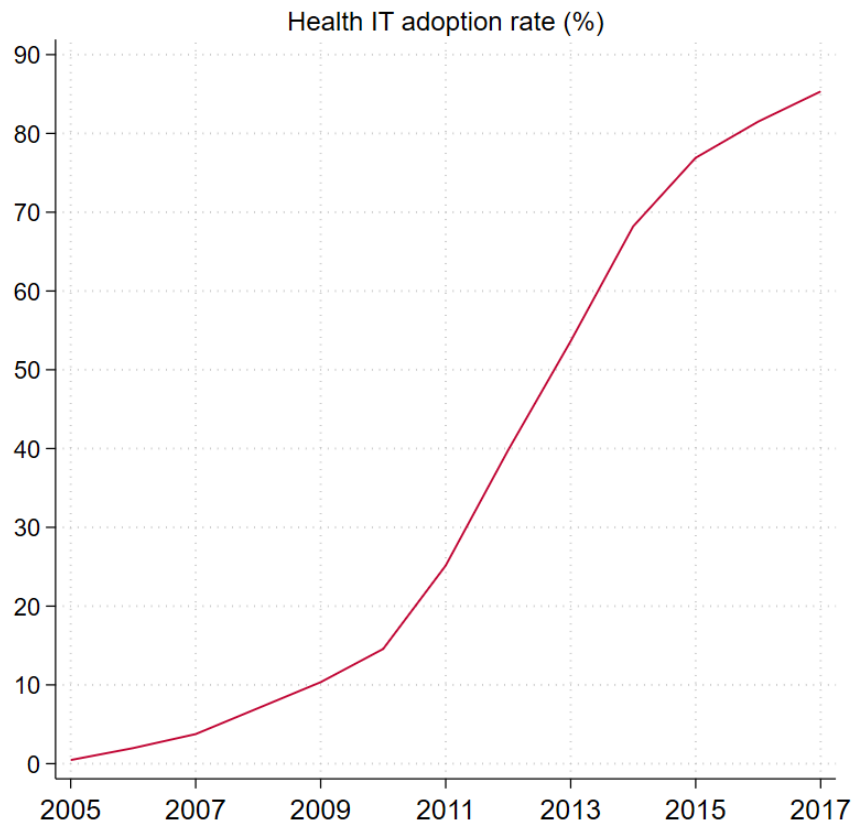
Boston University School of Law

Rising US HIT investments enables automation

- The US allocated over \$30 billion to health IT adoption (e.g., EHR)
 - Health Information Technology for Economic and Clinical Health (HITECH) Act and indirect provisions in Affordable Care Act (ACA)
- Extensive research on HIT impact on quality and cost
- But: limited understanding of **HIT** impact on **physician labor density**
 - Focus: **Computerized Provider Order Entry (CPOE)**, an advanced HIT and the key component of HITECH Meaningful Use (MU) criteria
 - E-enter treatment orders, integrate patient info and clinical guidelines, and communicate with pharmacy, laboratory, and other departments
 - In contrast, basic HIT systems, e.g., clinical decision systems, store patient data in digital database and provide clinical treatment support
- Research question: How does **advanced HIT adoption** affect **aggregate hospital-based (HB) physician density** at the county level?
 - # physicians **per 100k** population, **financial** measures, patient **care volume**

Institutional context and HIT adoption trends

- Electronic health records (EHRs), the core component of HIT, were invented in 1970s but saw limited adoption until the 2000s. We focus on hospital HIT.
- 2010s: widespread hospital HIT adoption, driven by federal incentives
 - 2009 HITECH Act provides incentive payments for HIT meaningful use (MU); ACA indirect incentives (initiatives/provisions) require/facilitate HIT use.



HIT adoption measures: CPOE & its functionality

- Computerized Provider Order Entry (CPOE): advanced HIT, [automation](#)
 - Enable providers to [electronically enter](#) medication, laboratory, and radiology orders, often with built-in [clinical decision support](#) tools, core in [MU criteria](#)
 - Unlike other HIT component, CPOE directly [structures patient data](#) to [align with coding and billing requirements](#), esp. relevant when payment matters

Order Entry

Alarms ADT Layout HIPAA Help Close

Facility: ADL Goodsam Demo Unit: 01 Resident: CARSON, JOHN

Acct: 123510 MedRecNo: 123510 Room: Admit: 11-02-03 Disch: Male Age: 95 Wgt: 149 lbs PCP:

Rx	End Date	Status	Procedure	End Date	Status	Lab Type	End Date	Status	Rehab Type	End Date	Status
ASPIR-TRIN 325	11-10-07	ACTIVE	APPLY OINTMEI	11-10-07	ACTIVE	BLOOD	10-07-07	DRAFT	PHYSICAL THEF	10-04-07	D/C
COUMADIN 2.5	11-10-07	ACTIVE	CHANGE DRESS	11-10-07	DRAFT	URINE	09-30-07	ACTIVE			
LASIX 20 MG	11-10-07	ACTIVE	CHANGE DRESS	10-07-07	DRAFT						
ENEMA READY	11-04-07	ACTIVE	O2@2LM NC	09-30-07	ACTIVE						
BISACODYL 10	11-04-07	ACTIVE	APPLY OINTMEI	09-30-07	ACTIVE						
EGL MILK OF	11-04-07	ACTIVE	CHANGE DRESS	09-30-07	ACTIVE						
ZYPREXA 10 M	10-26-07	ACTIVE									
ASPIR-TRIN 325	10-06-07	DRAFT									
NORVASC 5 MC	09-30-07	DRAFT									
COUMADIN 6 MC	09-30-07	ACTIVE									
BUSPAR 5 MG	109-30-07	ACTIVE									
FOSAMAX 70 M	09-29-07	ACTIVE									
ZYPREXA ZYDI	09-24-07	D/C									
ZYPREXA ZYDI	09-23-07	D/C									
ZOFRAN 4 MG	108-16-07	ACTIVE									
SINEMET-25/10	09-16-07	ACTIVE									
LASIX 20 MG	109-06-07	ACTIVE									
ACETAMINOPHE	08-24-07	ACTIVE									
BL ZINC GLUCO	08-24-07	ACTIVE									

LABS

REHAB

DIETS

RADIOLOGY

ACTIVITIES

Date	Dx	Description	Allergies	09-10-07	07-28-07	N/A	N/A	N/A
09-04-2007	332.	PARKINSON'S DISEASE*		Temp	99			
09-04-2007	331.0	ALZHEIMER'S DISEASE		Pulse	78			
				Resp	14			
				SBP / DBP	121 / 76	126 / 82		
				Weight	149			

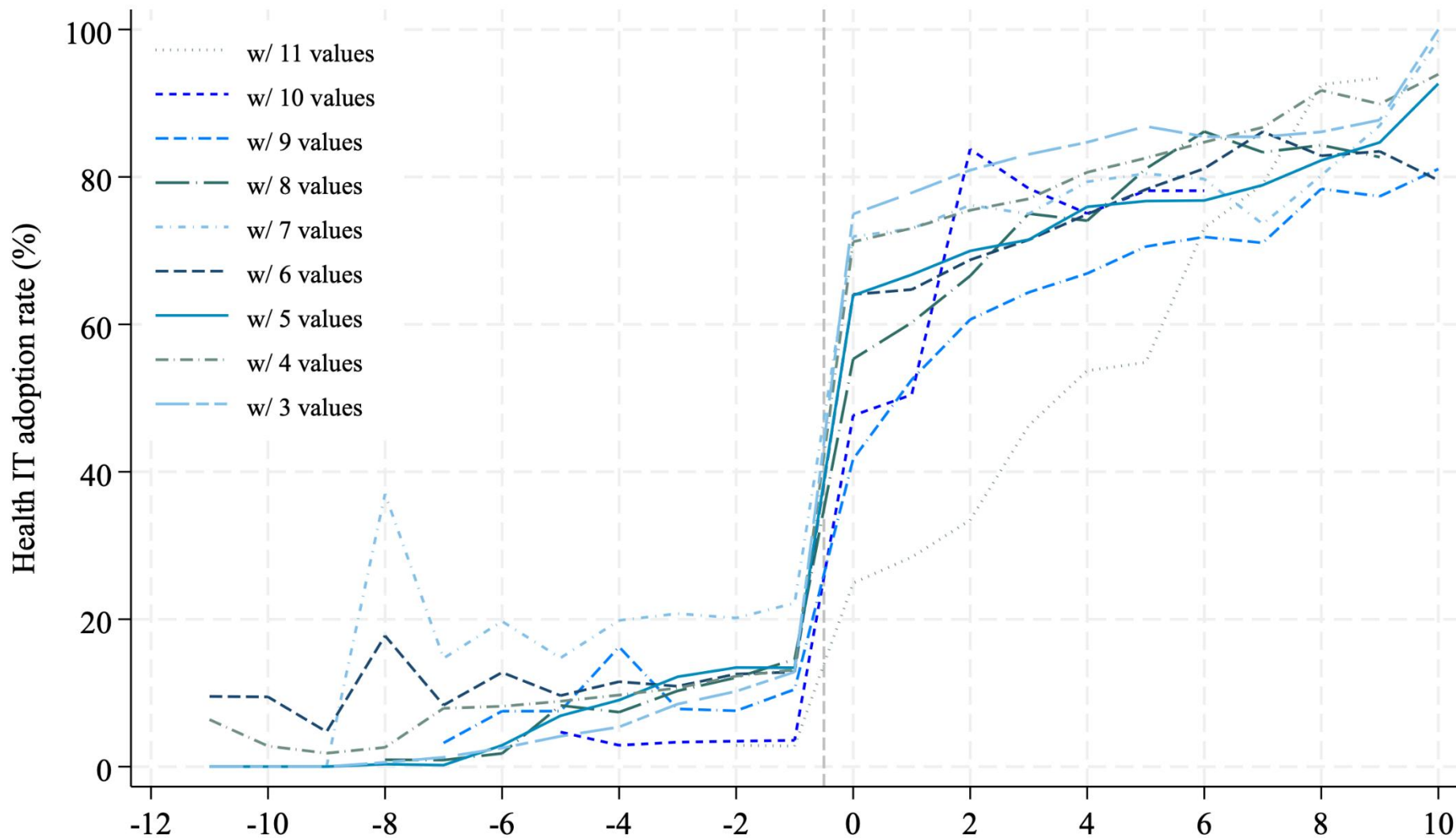
Virtual Body Care Plan Progress Notes ADLs Visits Notifications Summary

Data Construction

- **Physician supply data:** Area Health Resources Files (AHRF)
 - Hospital-based (HB) physicians: non-federal MDs under hospital contracts providing direct patient care, measured as rate per 100k county population
 - **Specialty categorization:** medical, surgical, general practitioners, others
- **Health IT adoption data:** Healthcare Information and Management Systems Society (HIMSS) Analytics Database
 - Records basic and advanced HIT adoption by hospitals, including CPOE
 - County-level adoption rate weighted by capacity (hospital beds)
- **Medicare provider and service data:** CMS Medicare physician & other practitioners by provider and service dataset (available starting 2013)
- **Market structure and profit data:** Healthcare Provider Cost Reporting Information System (HCRIS), CMS; calculate hospital HHI and profit
- **Sample:** ~2,394 US counties during 2006-2018 outcome years matched with one-year lagged HIT adoption (excl. 2009 b/c missing AHRF data)

HIT treatment measure: sharp jumps in HIT rates

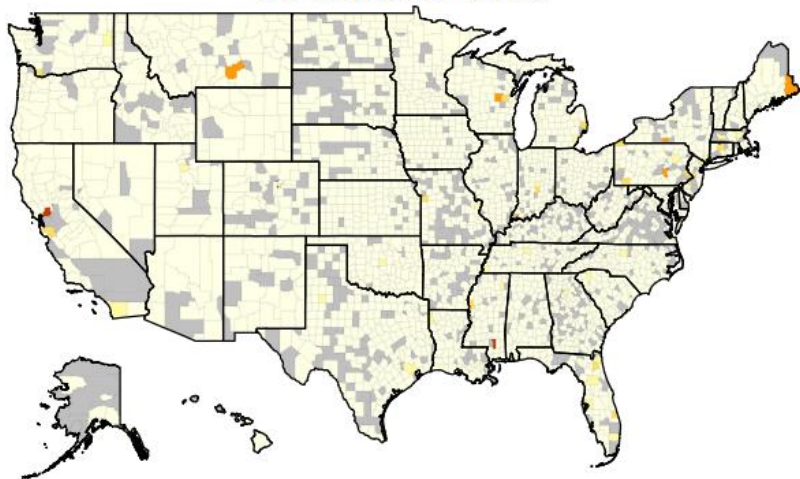
- Define treatment/control group à la the strategy in East et al. (AER 2023)
- Identify the sharpest jump in year τ (roughly = if $0 \rightarrow +$, or $\uparrow 100+\%$ at τ)



Geographic variation in HIT and physician supply

- Outcome periods 2006-2018 with one-year lagged HIT 2005-2017

HIT adoption rate (2005)



80 <= 100%

60 <= 80%

40 <= 60%

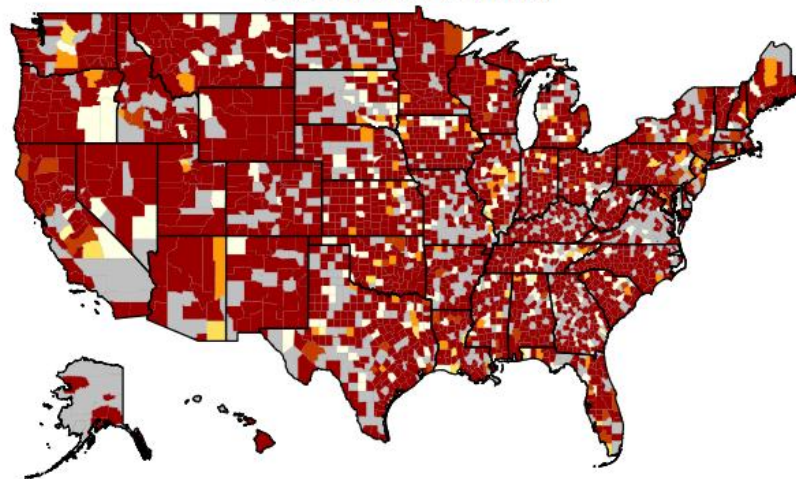
20 <= 40%

0 <= 20%

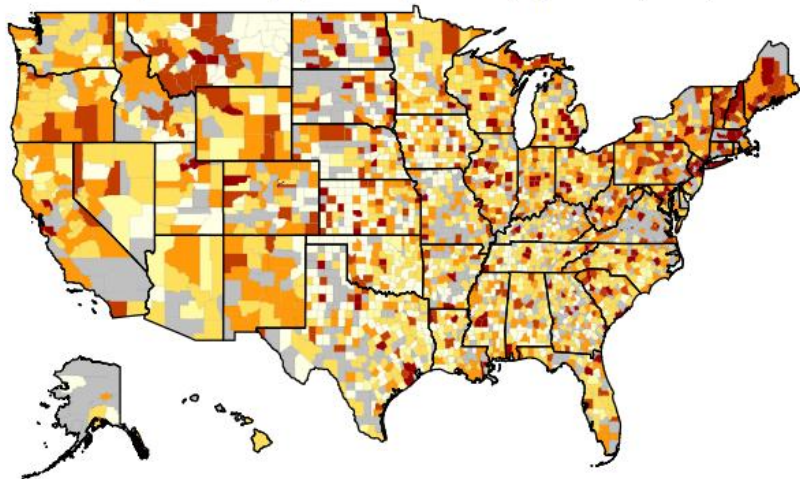
0

NA

HIT adoption rate (2017)



hospital-based physicians per 100k population (2006)



>40.23

21.49 <= 40.23

10.94 <= 21.49

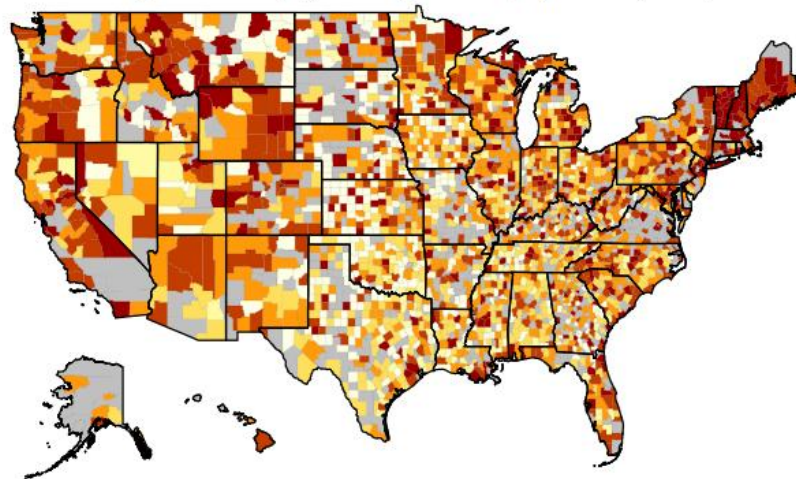
4.07 <= 10.94

0 <= 4.07

0

NA

hospital-based physicians per 100k population (2018)



Empirical strategy: dynamic DiD & event studies

- Staggered DiD design exploits the variation in the timing of the sharp increase in county HIT adoption (De Chaisemartin & d'Haultfoeuille, 2024)

$$Y_{it} = \beta_0 + \beta_1 Treat_{i(t-1)} + \gamma X_{it} + \lambda_{st} + \delta_i + \delta_t + \varepsilon_{it}$$

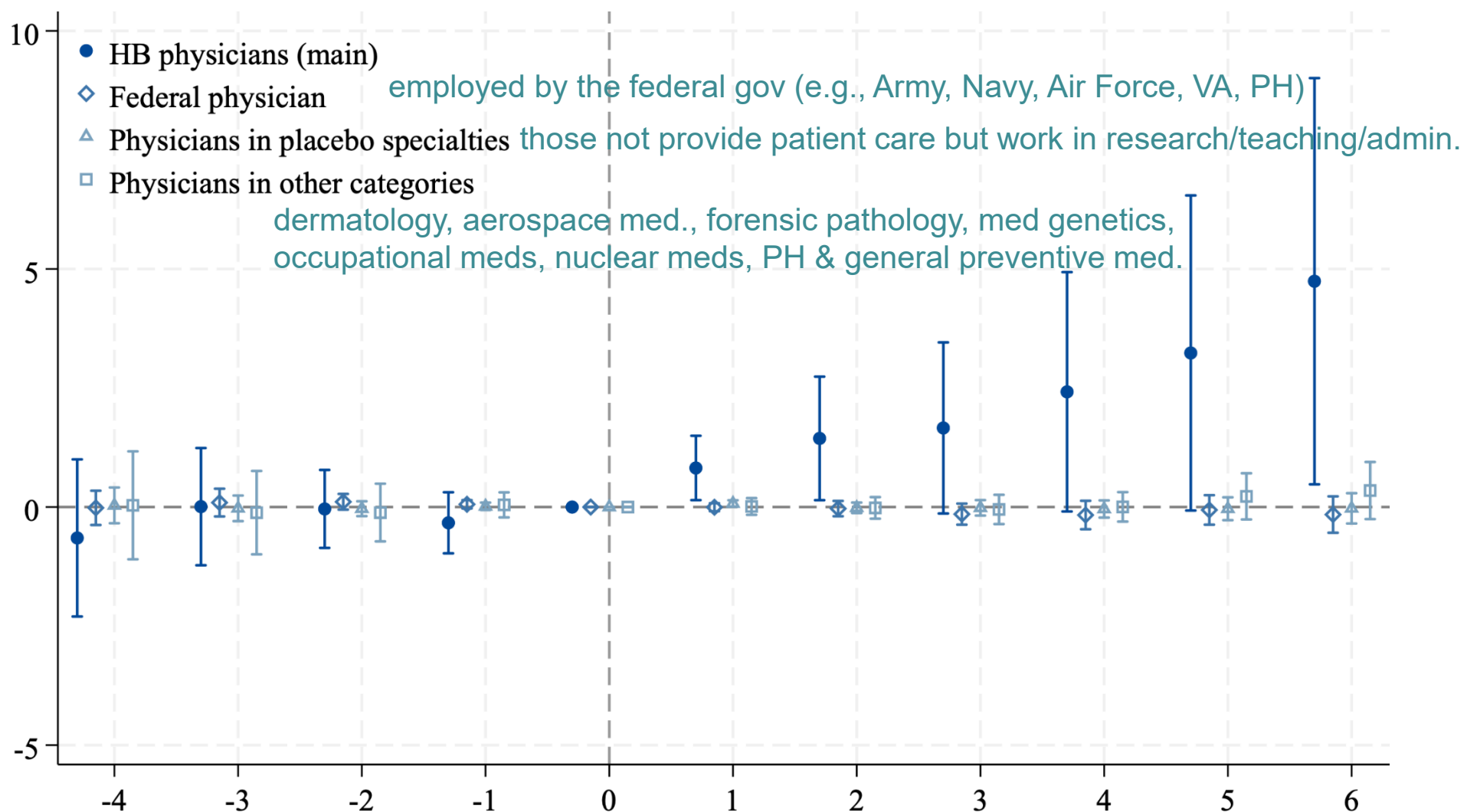
- Y_{it} : physician density in t matched with one-year lagged HIT treatment ($t-1$)
 - X_{it} : county-year observables controls: per-capita income, unemployment rate, %pop(65+), %Black, pop density, HHI (hospital), 2005-valued {Medicare advantage penetration rate, % college+ educated} $\times$ linear time trends
 - λ_{st} : state-specific time trends, where s denotes the state of county i
 - δ_i & δ_t : county and year fixed effects; ε_{it} clustered at the state level
 - Regressions are weighted by county-year level population
- Event study specification:

$$Y_{it} = \beta_0 + \sum_{\tau=-4, \tau \neq 0}^6 \beta_1^\tau \times 1\{t = \tau\} EverTreated_i + \gamma X_{it} + \lambda_{st} + \delta_i + \delta_t + \varepsilon_{it}$$

- Reference period $t=0$, treatment occurred in year 1 (as in dCDH 2024 setup)

HIT diffusion & HB physician: main results

- Main outcome: HB physicians per 100k pop. vs. three falsification groups



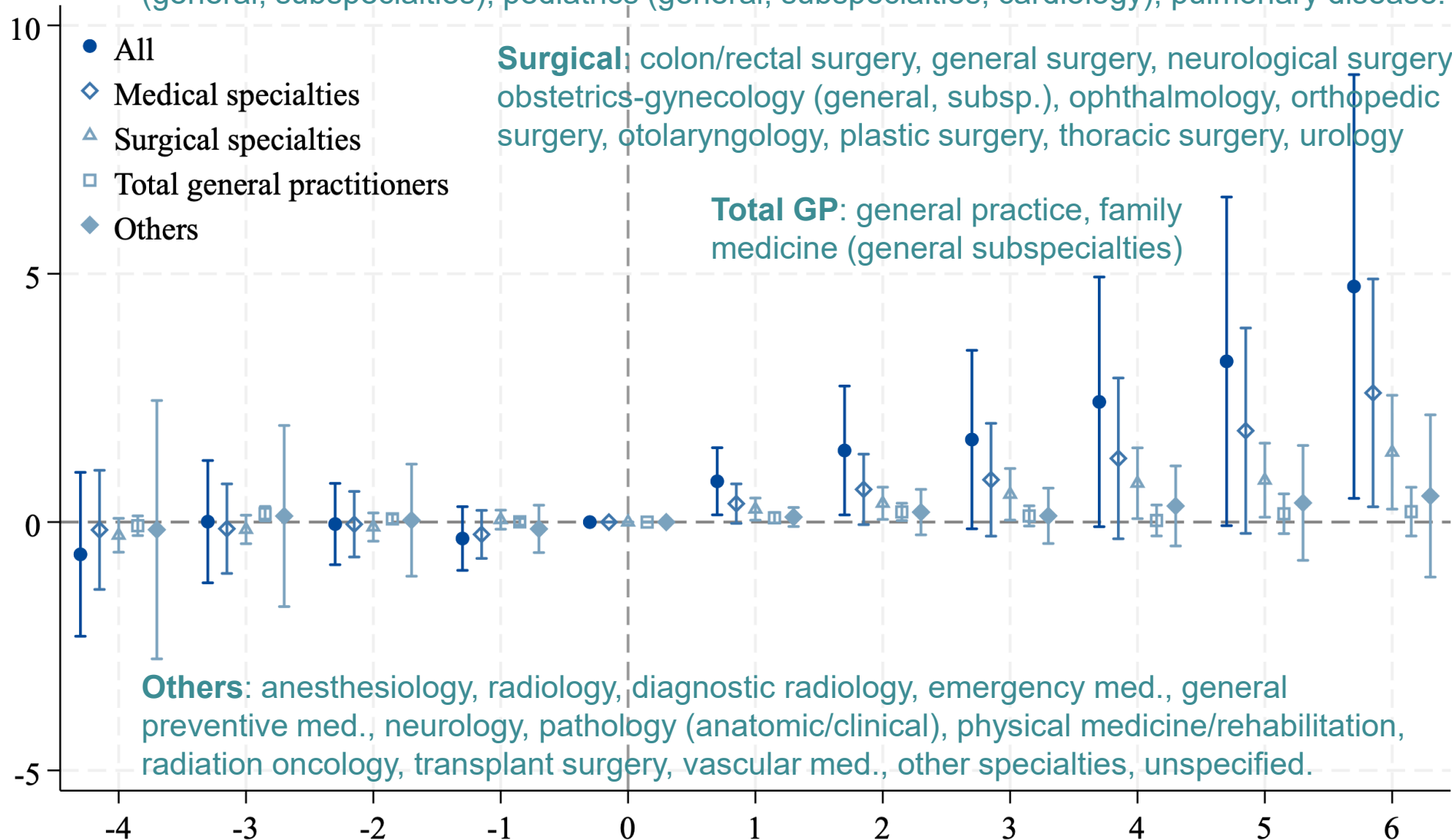
Heterogeneity cross-specialty within HB physicians

Medical: allergy and immunology, cardiovascular disease, gastroenterology, internal medicine (general, subspecialties), pediatrics (general, subspecialties, cardiology), pulmonary disease.

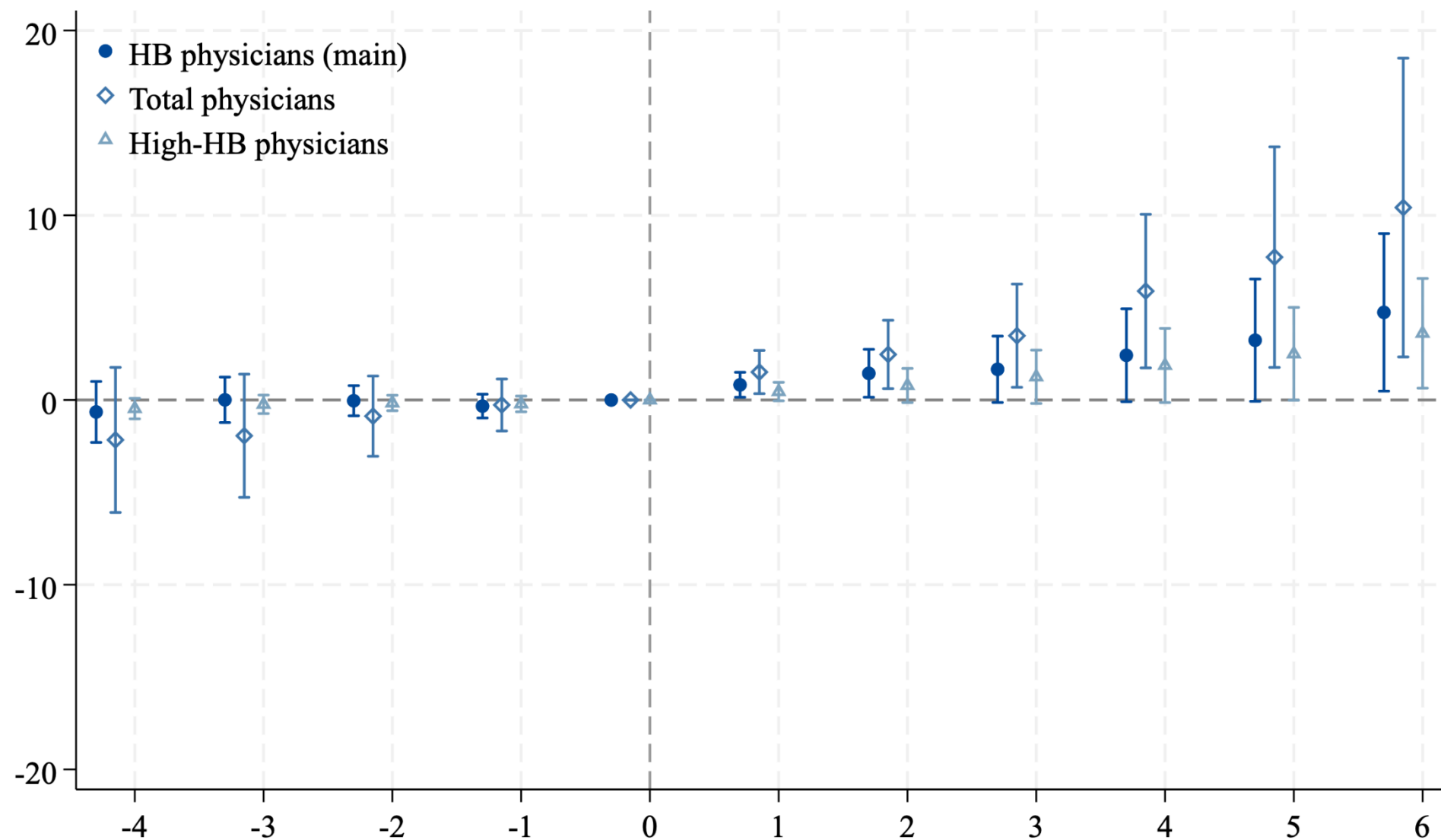
Surgical: colon/rectal surgery, general surgery, neurological surgery, obstetrics-gynecology (general, subsp.), ophthalmology, orthopedic surgery, otolaryngology, plastic surgery, thoracic surgery, urology

Total GP: general practice, family medicine (general subspecialties)

Others: anesthesiology, radiology, diagnostic radiology, emergency med., general preventive med., neurology, pathology (anatomic/clinical), physical medicine/rehabilitation, radiation oncology, transplant surgery, vascular med., other specialties, unspecified.

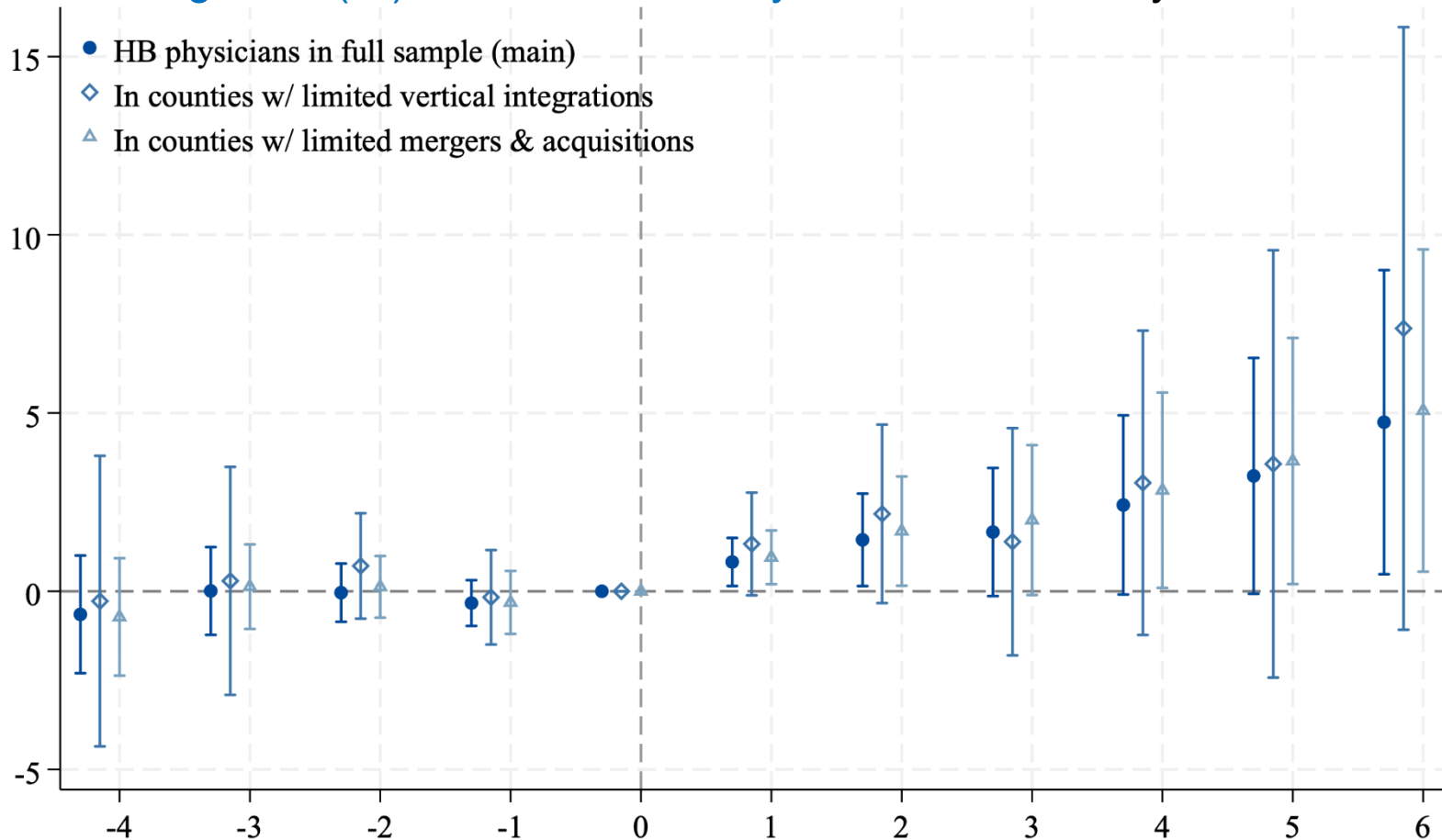


Expanded firm boundary or higher total doc density?



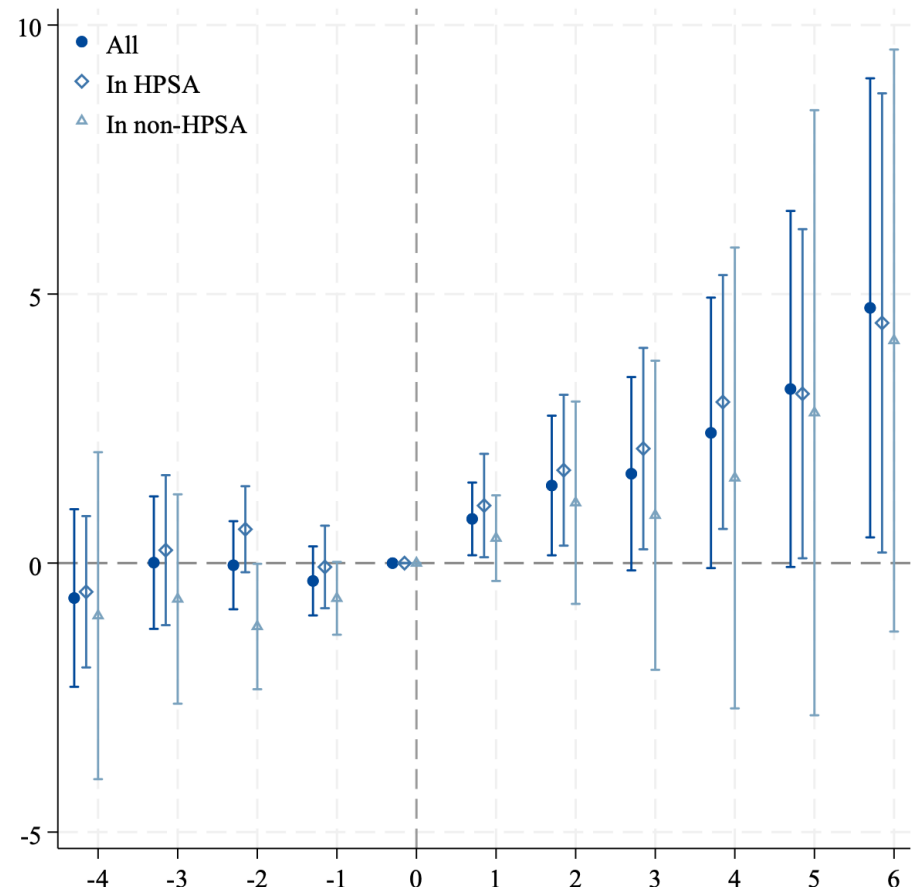
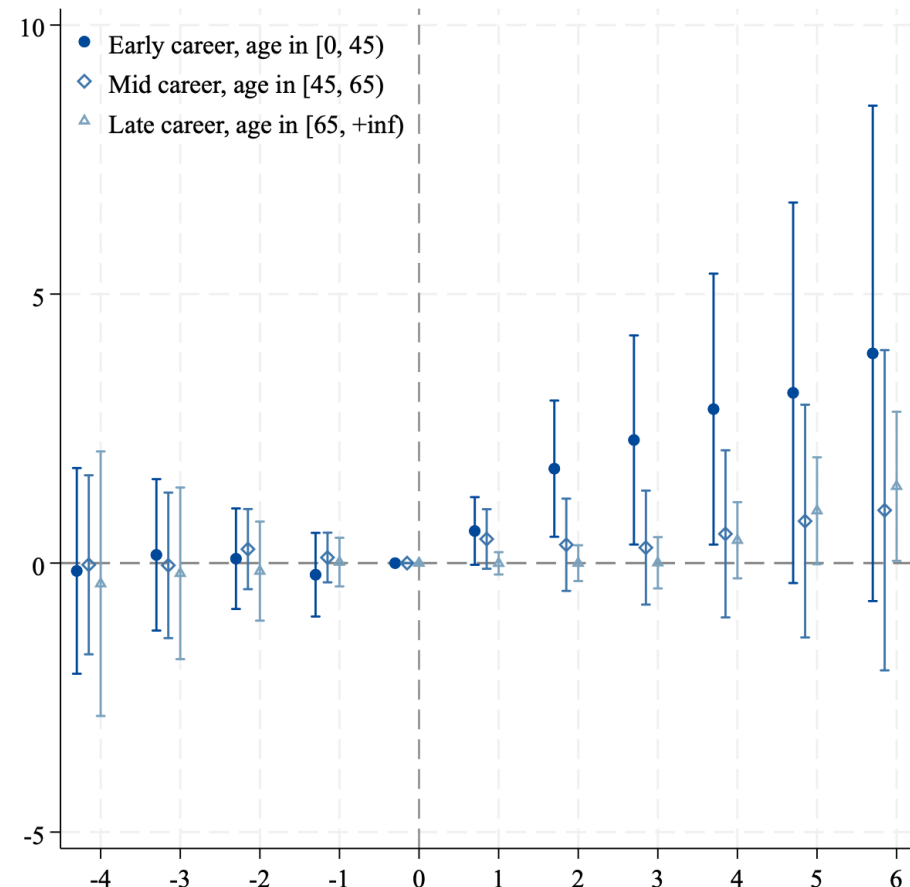
HIT Diffusion or Physician Practice Consolidation?

- Health systems acquire physician practices to get more patients (Baker et al., 2016)
- Systems, w greater financial & tech capacities, often adopt HIT early (Agha 2014)
- We identify health system that experienced a **merger & acquisition (M&A)**, if the **# hospitals within the system increased** in a given year compared to the previous year.
- a **vertical integration (VI)**, if the **# ambulatory care units** in the system increased...



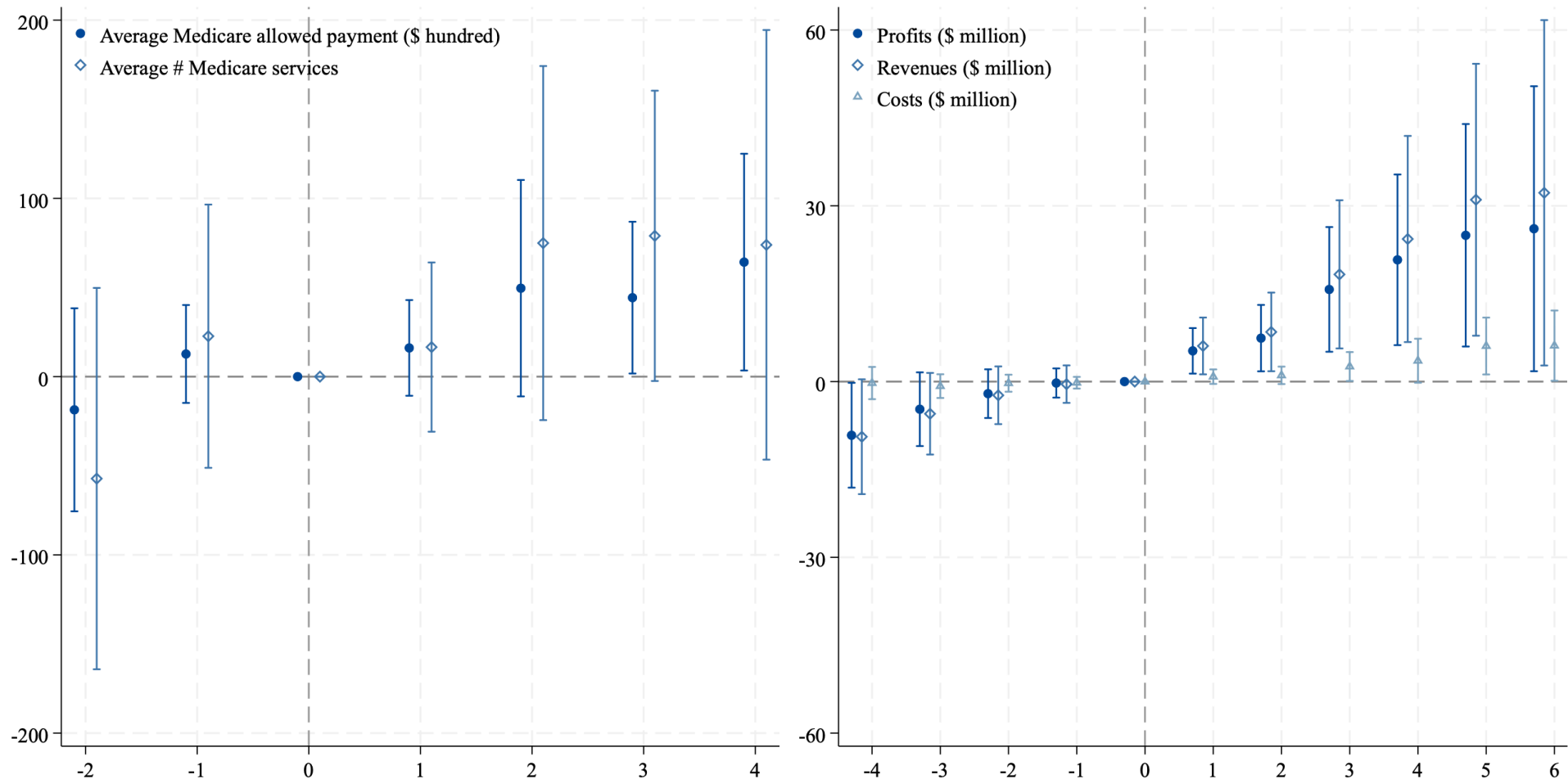
Heterogeneity by Physician Age & Shortage Status

- Early-career (under 45), mid-career (45-64), late-career (65+), total docs
- by Health Professional Shortage Area (HPSA) designation status
 - A county is ever-HPSA if at least part of the county was ever designated as having a shortage of primary care practitioners.



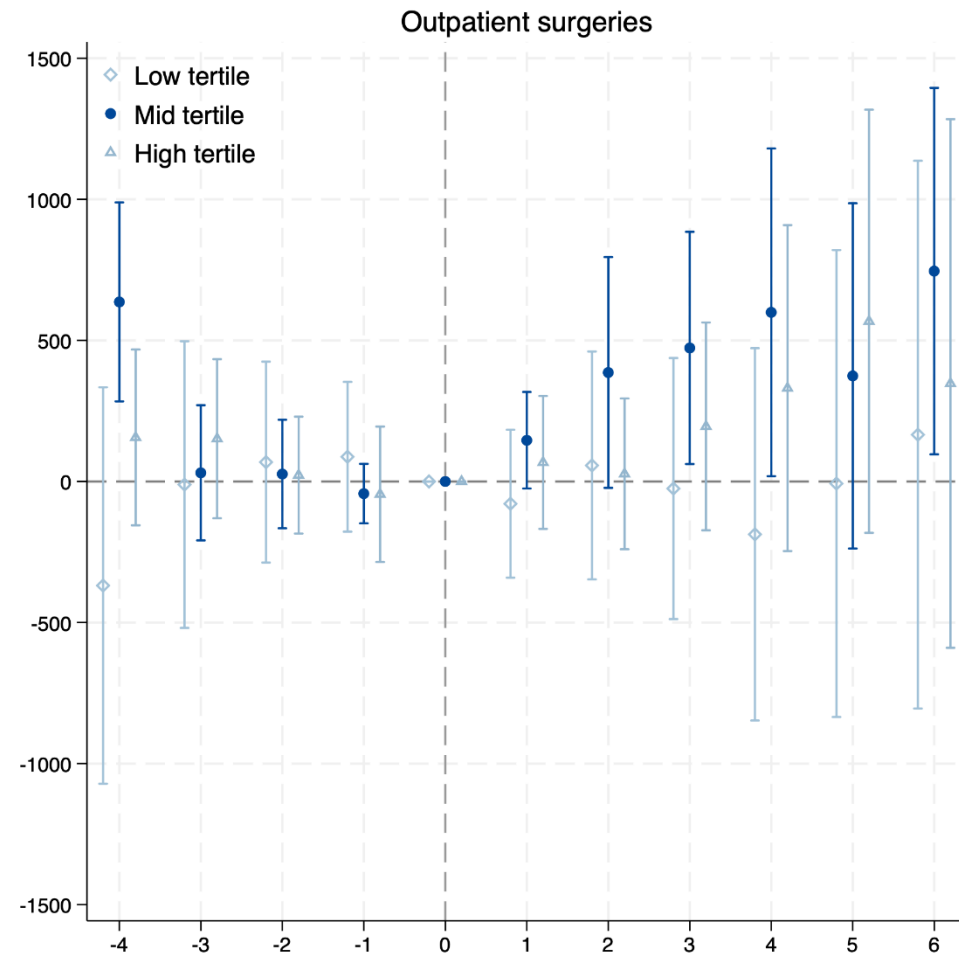
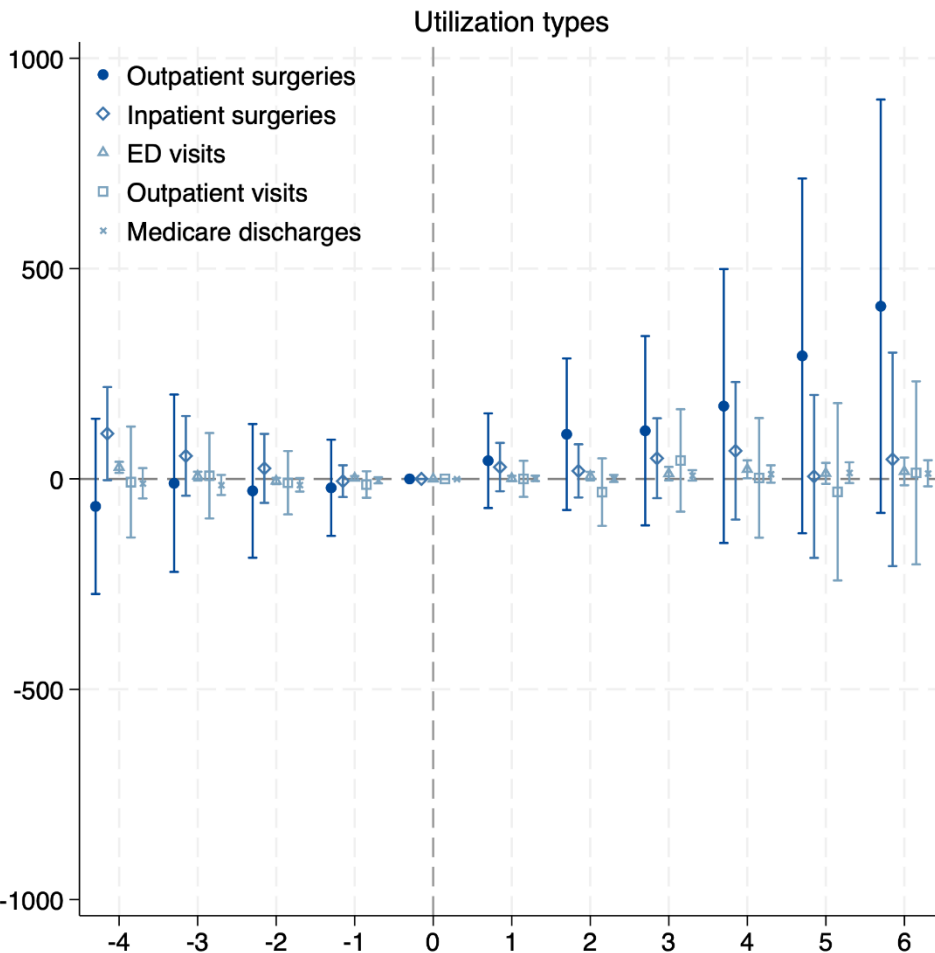
Potential Mechanism: the Financial Impact of HIT

- We use proxies for physician/hospital's performance:
 - [physician] avg Medicare-allowed amounts per HB physician, avg # of Medicare services per HB physician
 - [hospital] average hospital profit, revenue, and cost within each county



Heterogeneity by utilization: doc supply & care vol.

- Whether responses to HIT diffusion vary by **pre-period local care utilization levels**? And how this, in turn, influences **subsequent service use**?



Robustness checks

- Alternative **subsamples**

subsample

- Counties with ≤ 5 distinct adoption rates
- Counties that never had zero HB physicians
- Balanced panel with ≥ 4 periods before and after treatment

- Alternative model **specifications**

specifications

- Models without state-specific time trends
- Models without county-year level controls

- Alternative staggered DiD **estimators**

estimators

- Borusyal et al., 2024; Callaway & Sant'Anna, 2021; Sun & Abraham, 2021

- Alternative HIT **measures**

measures

- CPOE or physician documentation (PD), another advanced HIT

Limitation & future work

Conclusion & Discussion

- We find that aggregate HB physician density (county-level) **increases** after rapid adoption of advanced HIT, with effects concentrated among
 - More increases in medical/surgical specialties: tied with data & scheduling
 - Early-career physicians: age ≤ 45 , more movable and tech savvy
 - Health professional shortage areas (HPSA): places w excess demand to care
- Mechanism: treated counties showed better financial performance
 - Higher (per physician) avg Medicare allowed payments and services provided
 - Increased hospital profits, and greater outpatient surgical volumes
- Policy and managerial implications
 - Strategic HIT investment can attract (/retain) physicians to underserved areas
 - HIT (automation) appears to complement physician labor rather than substitute
 - Health tech diffusion may help reduce geographic disparity in care access

Health IT Diffusion and Physician Labor Supply



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Research Question:

Q: How does advanced HIT adoption affect aggregate hospital-based physician density at the county level?

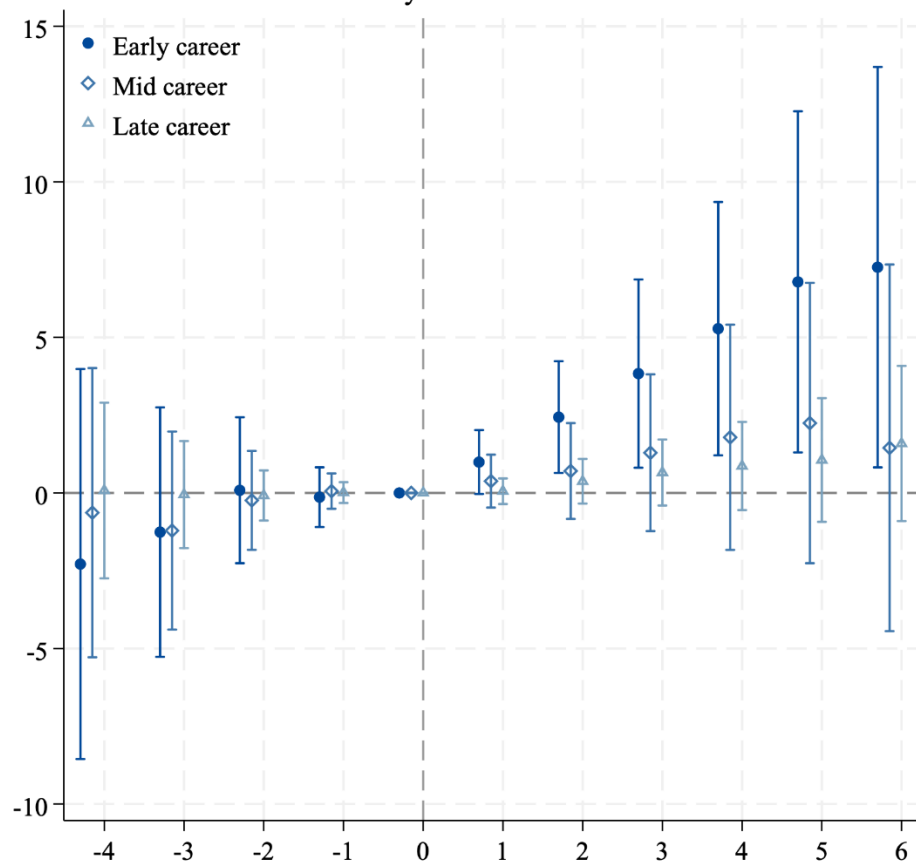
- Intuitively, aggregate physician rate can go either way (empirical Qs):
 - More: “Time to hire more physicians to take full advantage of HIT!”
 - Less: “Time to automate and reduce docs (recover costs from HIT)!”
 - Same: “Docs come and go, on the net things may stay similar?”
- Exploit the county-level sharp adoption of advanced HIT (CPOE)
- Show how the plausibly exogenous HIT diffusion patterns affect county-level HB physician per 100k pop, the heterogeneity by specialties, physician career stage, physician shortage area status
- Test financial mechanism and downstream impact on care utilization

Conceptual Considerations

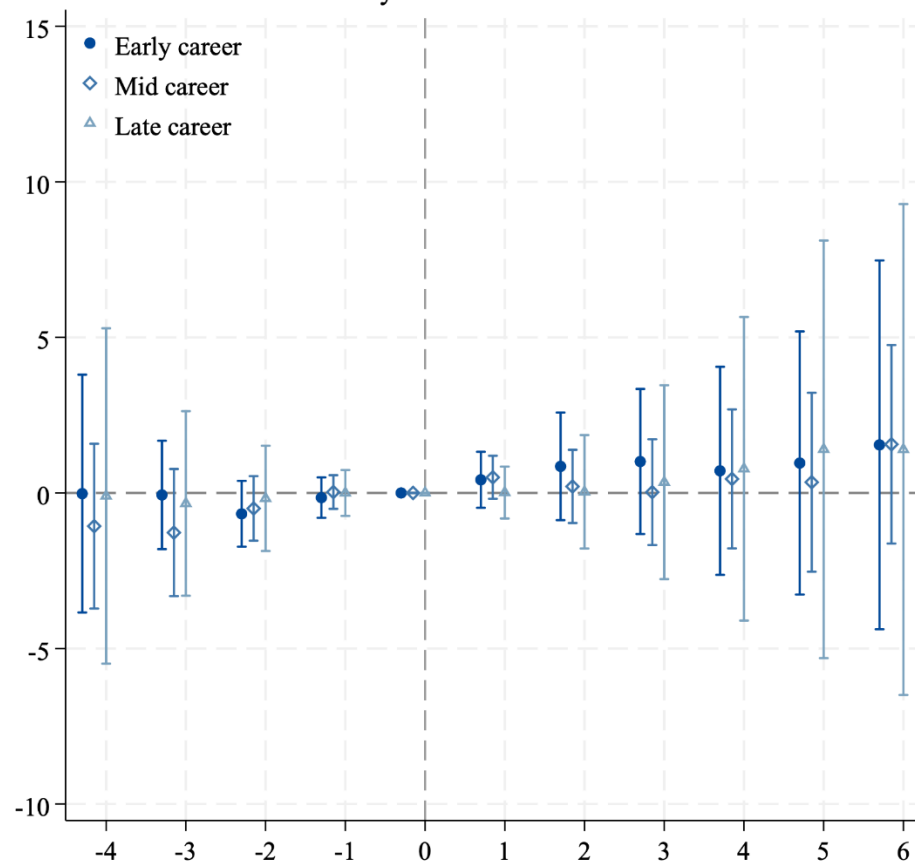
- Physician labor supply on the extensive margin is shaped by aggregating individual **location choices** and broader **market dynamics**.
- **Hospitals'** perspectives:
 - HIT investment can affect strategic hiring of docs to fully leverage technology
 - Esp in geo-areas with *moderate* but *growing* demand for medical services
 - If increase patient throughput, boost revenue -> can hire/retain more docs
- **Physicians'** perspectives
 - Response to HIT diffusion may vary by specialty, career stage, tech-savviness
 - HIT can be a pull factor for early-career physicians adapted to digital tech
- **Patients'** perspectives
 - HIT can make access to services more convenient with faster scheduling
 - can seek elective procedures earlier and easier (limited sorting within country)

Heterogeneity by career stage and HPSA status

Physicians in HPSA

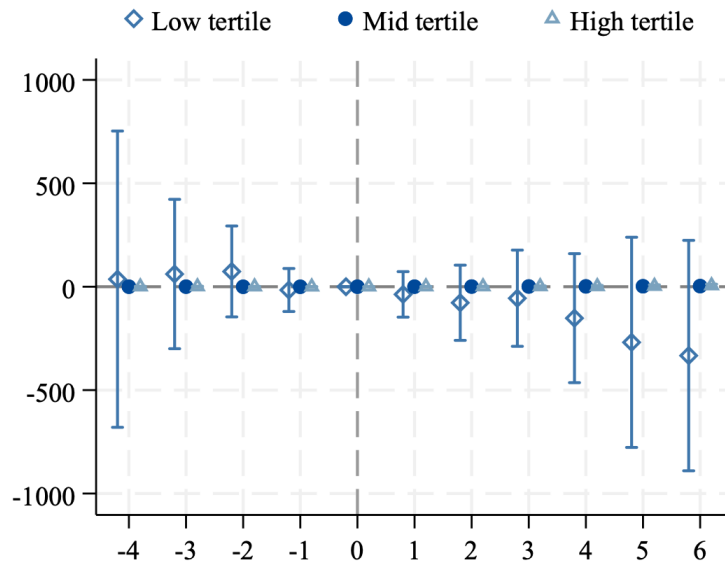


Physicians in non-HPSA

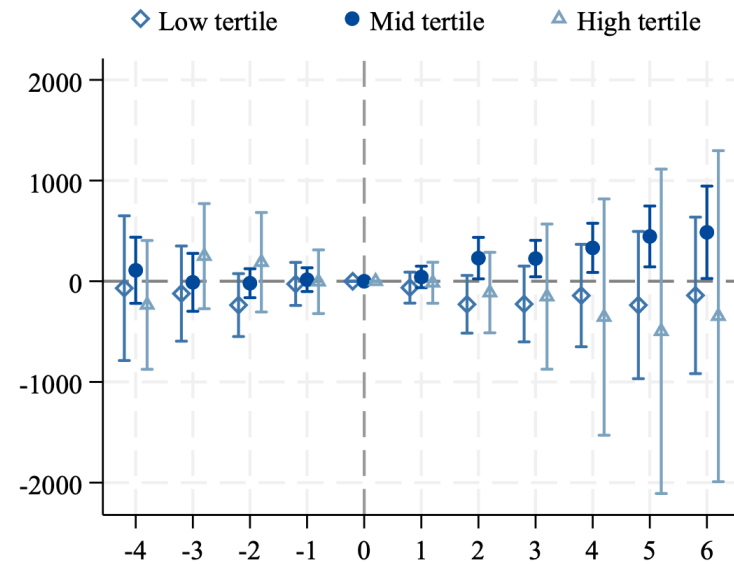


Heterogeneity by other utilization measures

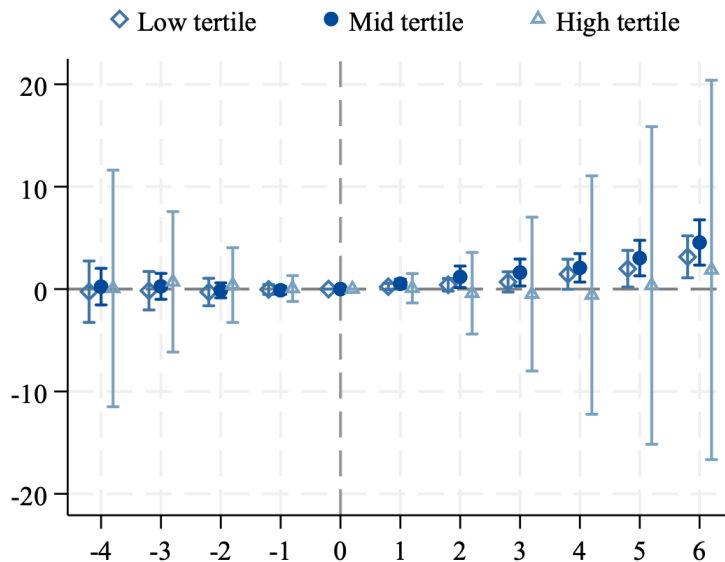
HB physicians by outpatient visit volume



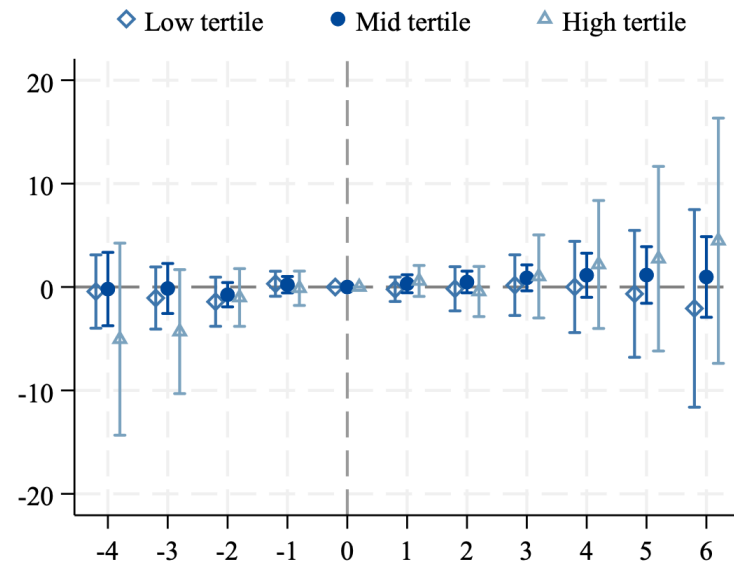
Outpatient visits by outpatient visit volume



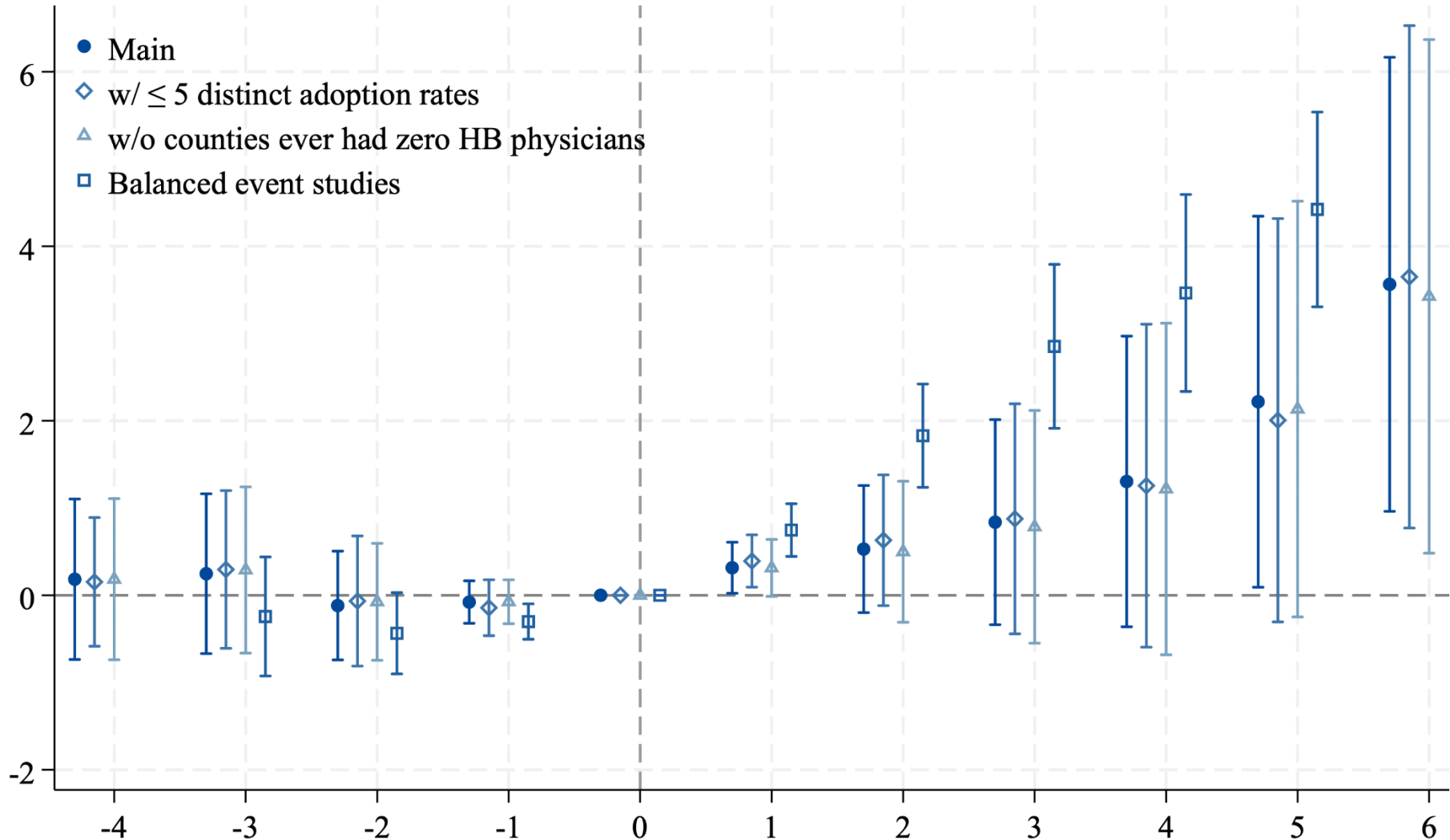
HB physicians by Medicare discharge volume



Inpatient discharges by Medicare discharge volume

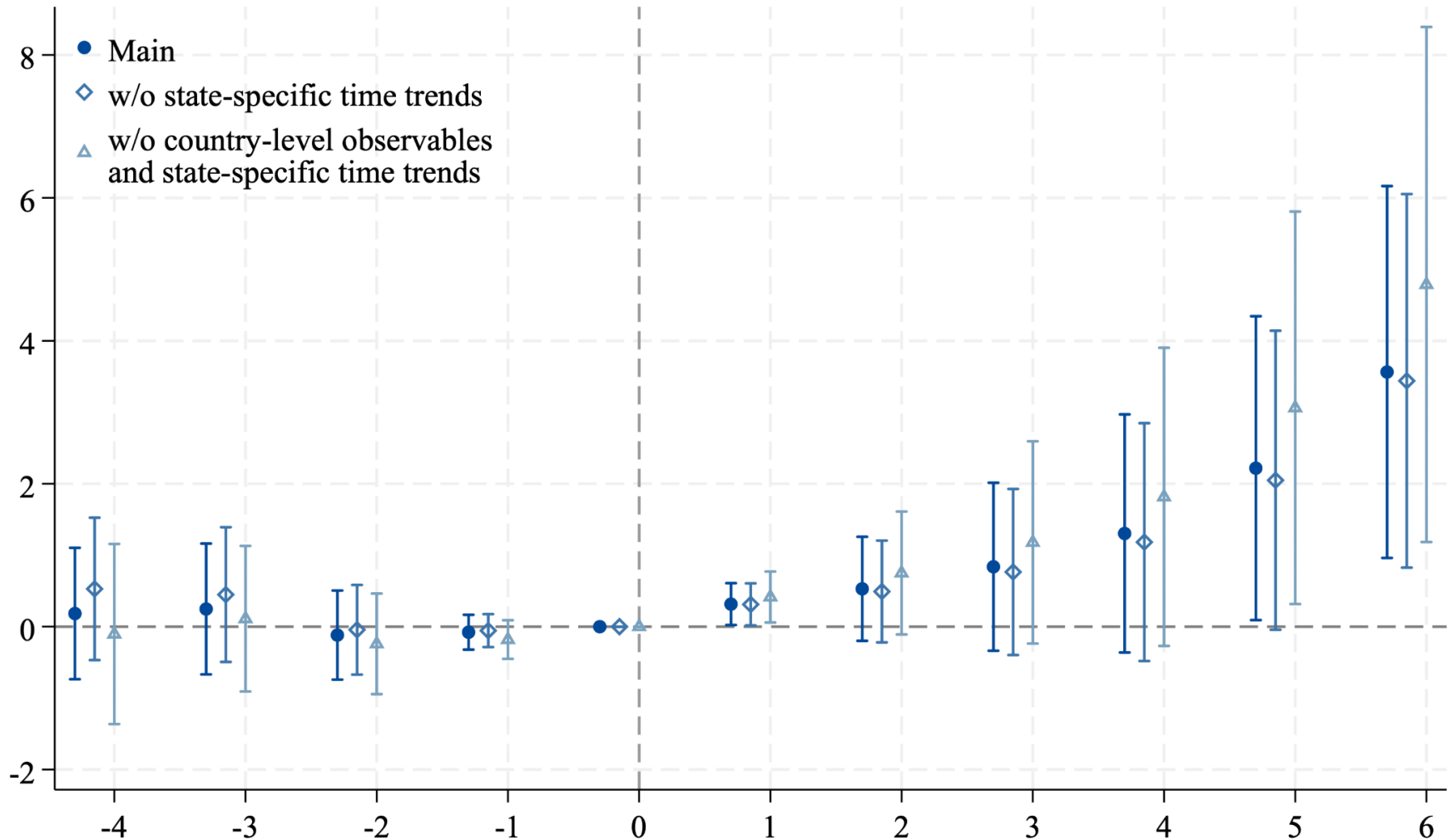


Robustness check: subsample sensitivity checks



Robustness check: specification checks

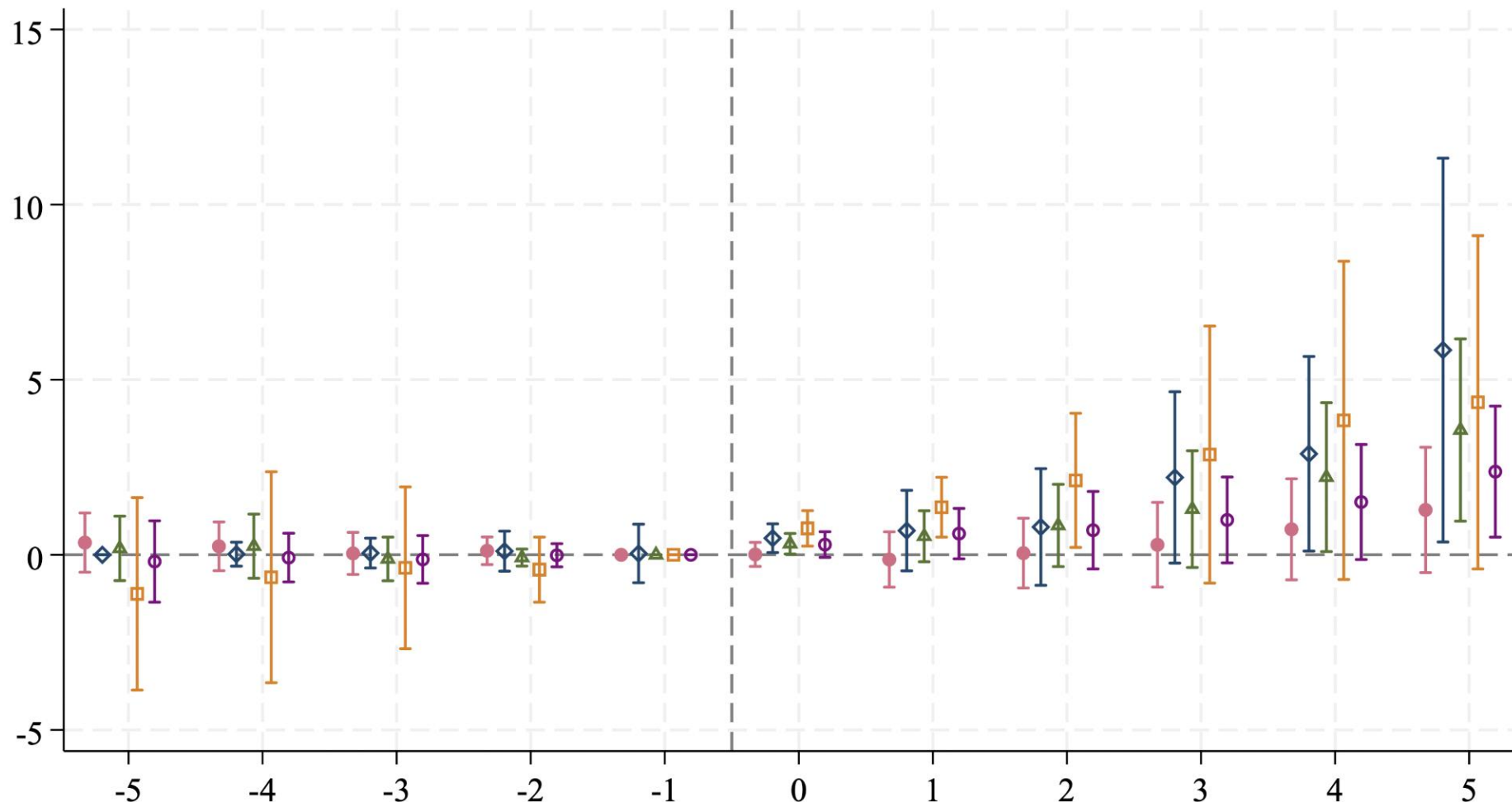
- Note the main specification always include state-specific time trends and county-level (including county-year level) observables.



Alternative DiD estimators for the main outcome

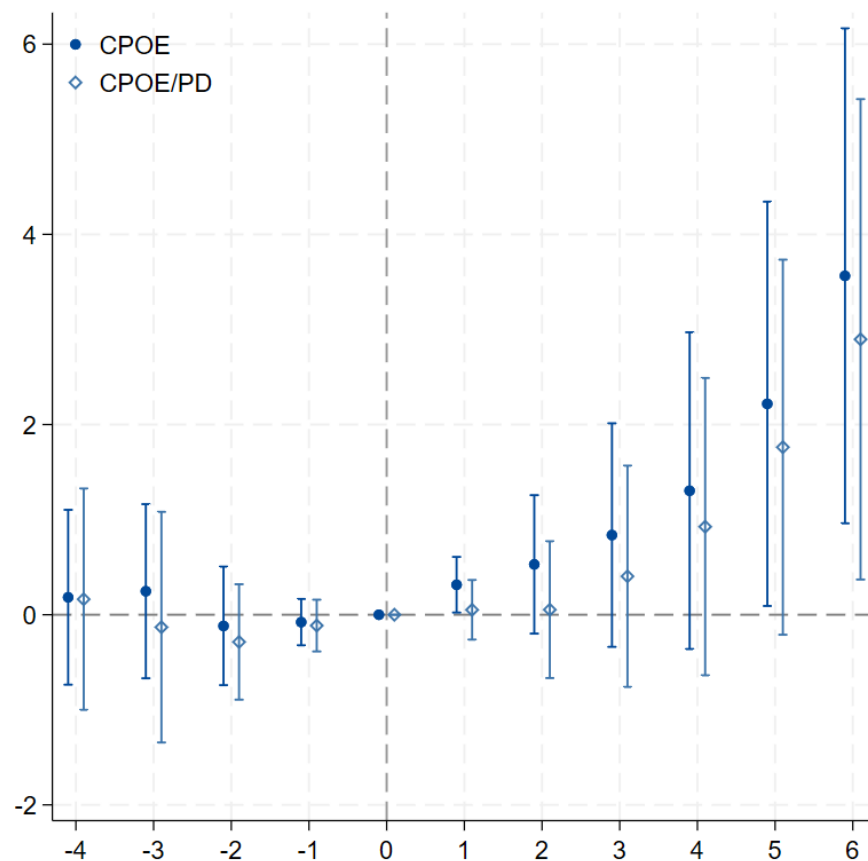
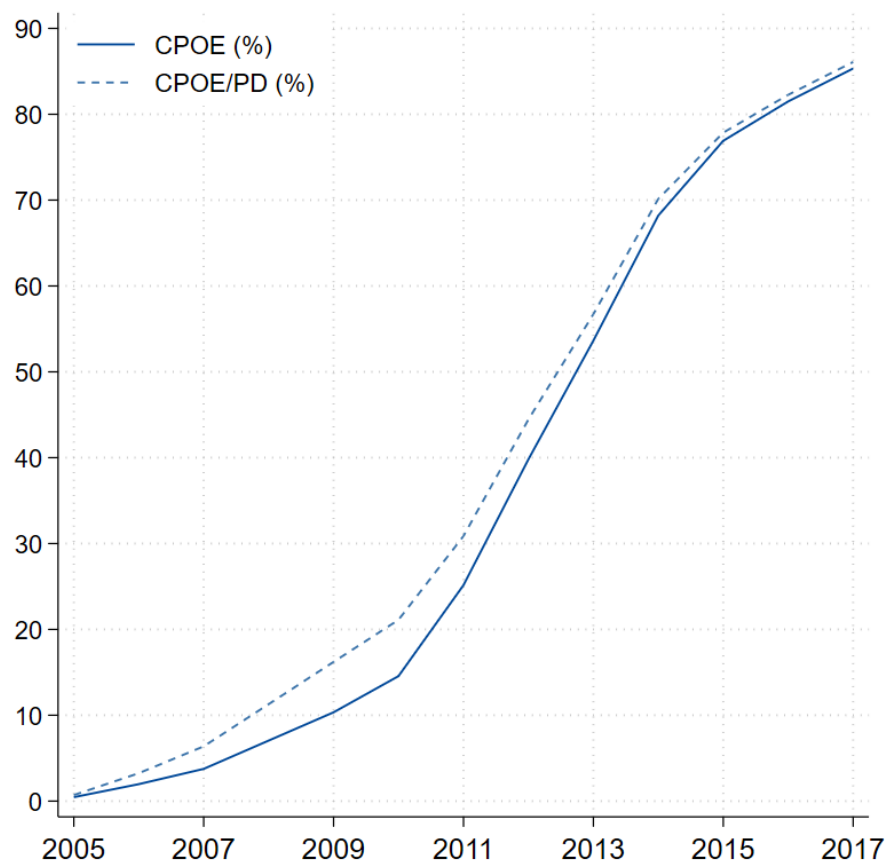
- All estimators use $t=-1$ as the reference except Borusyak et al., which uses the earliest period as the baseline period.

[back](#)



Alternative HIT measures: CPOE/PD adoption

- Alt. HIT adoption measures: county-level sharp adoption of either CPOE or physician documentation (PD) – another advanced HIT component (noted in Dranove et al., 2014) but not core for HITECH act MU criteria.



Limitations and Future Work

- Analyses based on *aggregate* data
 - ☺ mitigate concerns about patient sorting (limited sorting at county level)
 - ☹ limit our ability to disentangle changes at the physician level (e.g., replacing retiring physicians w junior docs, PCP vs specialists, roles...)
- Focus on *digital* tech but not fully capture *physical* med tech (data limit)
 - comprehensive controls mitigate concerns that other tech drive the results
- Cannot distinguish more nuanced *source* of responses
 - domestic vs international physicians, initial vs lateral practice location choices
- Not directly link with downstream *health outcomes*: HIT or docs?
 - HIT is one element within broader, team-based care delivery process
 - We demonstrate there can be direct and indirect effects via physician supply

Literature and Contribution

- **The impact of Health IT:** new evidence on how HIT automation affect aggregate physician labor supply (high-skilled labor), and how the effects vary by specialty, career stage, shortage area status, and care utilization.
(Miller & Tucker, 2011; Agha, 2014; Dranove et al., 2014; McCullough et al., 2016; Freedman et al., 2018; Lu et al., 2018; Wang, 2021; Wang & Bloch, 2023; Lin & Olson, 2024; Glenn, 2024)
- **Physician labor supply:** new technological angle that combines market-driven incentives and policy-led changes in the past decades, and study how digital technologies affect physicians, hospitals, and care provisions.
(Baker & Royalty, 2000; Garthwaite, 2012; Buchmueller et al., 2016; Falcettoni, 2018; Kulka & McWeeny, 2018; Neprash et al., 2018; Huh, 2021; Curto & Bhole, 2022; Huh & Lin 2024; Khoury et al., 2024)
- **Policy implications** and relevant insights for managerial practices. HIT investment can improve workforce efficiency and expand access to care (reduce disparity in access) while sustaining providers' financial health.

Summary statistics in pre-treatment period

	Overall	Never-treated	Ever-treated
HB physician counts per 100k population	10.27 (11.6)	10.30 (12.8)	10.27 (11.5)
Health IT adoption rate (%)	1.46 (5.39)	0 (0)	1.64 (5.69)
% 65+ population	15.7 (4.07)	17.2 (4.35)	15.5 (4.00)
% black	9.10 (14.4)	9.44 (15.6)	9.05 (14.2)
Population density	239 (1,722)	107 (520)	256 (1,815)
Income per capita	32,356 (8,325)	34,872 (10,304)	32,046 (7,996)
Unemployment rate (%)	6.65 (2.29)	6.65 (2.35)	6.65 (2.28)
Hospital HHI	0.80 (0.27)	0.92 (0.15)	0.78 (0.27)
Medicare Advantage penetration rate (%)	4.06 (7.73)	2.87 (5.58)	4.21 (7.95)
% above college degree	12.6 (5.7)	11.4 (5.1)	12.8 (5.7)
# counties	2,394	265	2,129